

PYTHON OOP BASIC NOTES (Classes, __init__, self, __str__)

1. What is a Class?

A class is a blueprint (template) used to create objects.

Example: class Student:

2. What is an Object?

An object is a real instance of a class.

Example: s1 = Student()

3. What is __init__()??

__init__() runs automatically when an object is created.

Used to store values inside the object.

Example: def __init__(self, name): self.name = name

4. What is self?

self means the current object.

Example: s1 = Student('Jashan') -> self = s1

Example: s2 = Student('Rahul') -> self = s2

5. Why do we write self.name?

self.name stores data inside the object.

Each object keeps its own data separately.

6. What is __str__()??

__str__() controls what prints when we use print(object).

Example: def __str__(self): return f'Name is {self.name}'

7. Important Rule

Class = blueprint

Object = real item created from class

`__init__` = stores values when object is created

`self` = current object